Gravity, Trade and Transport

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ECON 803, Fall 2026

Questions only. Restored from both parts of the SFU December 2024 final: the structural gravity argument and the Arabasta and Baltigo modeling case. Each question separates hand work from Codex and verification. Timing refers to the hand work. The original contraction request and the counterfactual income closure are corrected explicitly.

These are the same questions as the textbook. Begin with the hand calculations and state a solution plan before using tools. Report the economic assumptions, the hand seed, one scaling step and an exact verification command. Numerical output alone is not a proof.

Optional background: Anderson and van Wincoop (2003), Gravity with Gravitas

Problem 16.A1. CES trade, market clearing and the limits of a fixed point argument

Restored from the theory problem in the SFU Fall 2024 final examination. The original four stage argument and CES weighting are retained. The symmetry and fixed point claims are corrected. The following timing and points are for practice, not the original exam allocation.

Suggested time: 70 minutes. Total: 60 points for Part I.

Part I. By hand

There are $N \geq 2$ countries, indexed by i for origins and j for destinations. Country i owns a fixed quantity $Q_i > 0$ of its unique variety, has no other goods endowment, and has factory price $p_i > 0$. Its income is $y_i = p_i Q_i$. All varieties are traded competitively. Delivering one unit from i to j uses $t_{ij} \geq 1$ origin units, with $t_{ii} = 1$. Delivered prices are $p_{ij} = p_i t_{ij}$. There are no outside transfers or trade deficits. Destination j chooses nonnegative delivered quantities c_{ij} with budget $\sum_i p_i t_{ij} c_{ij} \leq y_j$ and utility $U_j(c) = [\sum_i \beta_i^{(1-\sigma)/\sigma} c_{ij}^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$, where $\beta_i > 0$ and $\sigma > 0$, $\sigma \neq 1$. For $0 < \sigma < 1$, define utility to be zero when any component is zero, its continuous boundary value. You may use the standard concavity of this CES aggregator. Bilateral expenditure is $x_{ij} = p_i t_{ij} c_{ij}$, including domestic expenditure when $i = j$. Let $Y = \sum_i y_i$ and $\theta_i = y_i / Y$.

(a) Derive the optimal bilateral quantities, bilateral expenditures and destination unit expenditure index P_j, step by step. Explain why the optimum exhausts the budget, is interior and is globally optimal. Your argument must preserve the displayed β_i weighting. [16]

(b) State physical market clearing, including iceberg use, and its nominal form. Use your expenditure formula to isolate $\beta_i p_i$ as a function of y_i, the trade costs and the destination indices. Explain why the income identity $y_i = p_i Q_i$ must still hold when interpreting the result as the original fixed supply economy. [10]

(c) Define outward multilateral resistance by $\Pi_i^{1-\sigma} = \sum_j \theta_j (t_{ij}/P_j)^{1-\sigma}$. Eliminate $\beta_i p_i$ to derive the structural gravity equation and the inward resistance system. Show the reciprocal scale freedom of the two resistance systems and state a normalization. Take logs to identify the exporter effect, importer effect and coefficient on $\log t_{ij}$. Distinguish the conditional index normalization from changing nominal currency units in the underlying economy. [16]

(d) Suppose $t_{ij} = t_{ji}$. You may use the following result without proof: for a strictly positive finite trade cost matrix and positive shares summing to one, the joint inward and outward resistance system has a positive index pair, and any two such pairs differ only by reciprocal scalar rescaling. Use this result to explain when a normalization with $\Pi_i = P_i$ is available, and why symmetry alone does not make every unnormalized pair equal.

For the rest of this part use exactly the original diagnostic values $N = 2$, $\sigma = 1/2$, $t_{ij} = 1$ and $\theta_i = 1/2$. After choosing the symmetric normalization, write the inward equations as $P = g(P)$. Determine whether g maps $\{P : P_i \geq 1\}$ into itself. Then use $K = [1/2, 2]^2$: prove that g maps K into itself and apply Brouwer to establish a fixed point. Determine all positive fixed points directly. Compute Dg at $(1, 1)$ and decide whether an induced matrix norm can be strictly below one there. Finally calculate two iterations from $P^{(0)} = (2, 2)$ and explain what they show about repeated substitution, existence and uniqueness. [18]

Part II. Codex and verification

Construct a synthetic economy with at least ten countries, positive β_i, Q_i, finite iceberg costs and $\sigma > 1$. Keep fixed physical supplies rather than silently treating both factory prices and nominal incomes as independent data. Solve for positive factory prices in log variables with one numeraire normalization. Compute incomes, delivered consumption, expenditure, both resistance systems and the physical shipment totals. Check every consumer budget, both row and column expenditure sums, all physical goods markets, positivity and the omitted redundant equation. Repeat under a change of currency units and verify how prices, incomes, nominal trade and physical quantities transform. Compare a normalized resistance iteration with a root solver conditional on the resulting income shares, align their index gauges before comparing them, and report full residuals as well as stopping criteria. For the synthetic fixed effect regression, require nonadditive bilateral variation in $\log t_{ij}$ and report the design rank after choosing a dummy normalization. Recover the theoretical cost coefficient. Reproduce the two country diagnostic separately and do not infer convergence from an equation having a unique solution.

Problem 16.A2. Arabasta and Baltigo: building the original gravity model of trade

Restored from Problem 2 of the SFU December 2024 final examination. The original six modeling tasks, Arabasta and Baltigo, unequal country incomes, transport technology and distance comparisons are retained. Part II is a new computational extension.

Part I. By hand

Suggested time: 75 minutes. Total: 60 points.

There are two countries, Arabasta a and Baltigo b, and one good from each origin. Quantity c_{ij} is consumption in destination j of origin i's good. Nominal income and expenditure are $y_a, y_b > 0$. Origin prices are $p_a, p_b > 0$. Domestic delivery has no resource loss, while bilateral trade has the symmetric iceberg factor $\tau \geq 1$: $p_{aa} = p_a$, $p_{bb} = p_b$, $p_{ab} = p_a\tau$ and $p_{ba} = p_b\tau$. Delivering one unit abroad requires τ units at origin. Bilateral expenditure is $x_{ij} = p_{ij}c_{ij}$ and origin revenue clearing requires $y_i = x_{ia} + x_{ib}$.

Use equal origin weights and common CES preferences. Write the substitution elasticity as $\sigma > 1$ and use $\sigma = 2$ for the closed form equilibrium calculations. The original key's power utility with exponent $1/2$ is ordinally equivalent to this case; its exponent is not the substitution elasticity.

Normalize $p_b = 1$. In Part I, y_a and y_b are fixed nominal incomes measured in these units. Origin supply quantities are not independently fixed: after solving the revenue equations they are implied by $Q_i = y_i/p_i$. State this conditional nominal income closure when interpreting a counterfactual. It is not a comparison that holds both nominal incomes and physical endowments fixed.

(a) Original task 1.1. Specify a one parameter CES utility family and Arabasta's budget. Explain which goods, prices and income enter this problem. [10]

(b) Original task 1.2. Write the Lagrangian and derive both consumption quantities in Arabasta. Relabel carefully to obtain Baltigo's demands. Give a global optimality argument and state where your solution is interior. [10]

(c) Original task 2.1. Write both countries' revenue clearing conditions. Specialize to $\sigma = 2$, impose $p_b = 1$, and solve explicitly for the unique positive origin price p_a as a function of y_a, y_b, τ. Reject any algebraic root that is not an admissible price, and check the case $\tau = 1$ directly. [10]

(d) Original task 2.2. Let $d \geq 0$ be distance. Propose a one parameter engineering transport technology $d = f(k)$ with nonincreasing returns in ship capital services k. Normalize the rental cost of these services to one in the origin good resource units used to define iceberg shipping. Derive the per delivered unit resource requirement $C(d)$ and set $\tau(d) = 1 + C(d)$. Verify $\tau(0) = 1$ and that larger distance increases the barrier. Explain why you must not add a second monetary transport bill to the country budget. The historical key's constant returns choice is admissible; a concave power technology is also admissible. [10]

(e) Original task 3.1. Eliminate the origin price from Arabasta's equilibrium import quantity c_{ba} and nominal imports x_{ba}. Derive the corresponding exports c_{ab}, x_{ab} and verify both budgets and both physical resource equations, including iceberg use. As a common numerical check, use $y_a = 5$, $y_b = 2$, $d = 1$ and the admissible technology $f(k) = \sqrt{k}$. This check is part of the existing

task, not an additional scored question. [10]

(f) Original task 3.2. Prove that doubling both nominal incomes doubles bilateral nominal trade in this closure. Then prove that increasing distance strictly reduces bilateral nominal imports, holding those incomes fixed. Separate the response of import quantities from nominal import values, and explain why this is not yet a welfare theorem for a fixed physical endowment economy. [10]

Part II. Codex and verification

This is a new extension, separate from the restored final question. Begin only after completing the hand arguments.

1. Use the numerical seed from part (e), fix the engineering parameter, and vary only distance over $d \geq 0$. Ask Codex to implement your analytical price, expenditure and quantity formulas, and independently solve the two revenue equations. Compare both methods. Test positivity, budget exhaustion, balanced bilateral expenditure, the domestic expenditure residuals, and every origin resource equation. Include zero distance without evaluating a formula that divides by zero. Plot bilateral expenditure and both implied origin supply quantities. Explain explicitly which physical quantity changes when nominal GDP is held fixed.

2. Optional different closure. Freeze physical endowments at the values you derived from the seed, and retain $p_b = 1$. Now income must be endogenous, $y_a = p_a Q_a$ and $y_b = Q_b$. Derive the positive price equation under this closure, establish uniqueness of its positive root, and solve it as distance varies. Verify the original consumer problems, physical market clearing and income accounting independently. Compare the two closures at the seed and away from it. Any utility comparison must use the same CES preference and the full fixed endowment closure, not interpret the fixed nominal GDP family as the same physical economy.